PRE- BOARD EXAMINATION (2024-25)

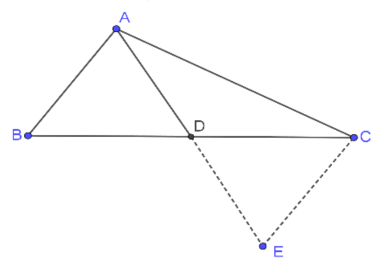
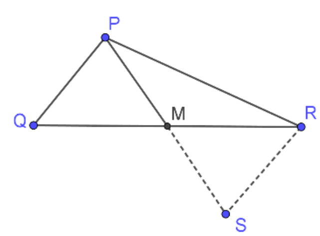
CLASS – X

TIME: 3 Hrs SUBJECT: MATHEMATICS STANDARD (041) M.M: 80

MARKING SCHEME

S.No.	SOLUTION	MARKS
SECTION A		
1	Option (d) none of these (answer- 3/2 and -1/3)	1
2	(a). 2:1	1
3	(c) 8 - 3n	1
4	(d) 150 deg	1
5	(b) -10	1
6	(d) 18cm	1
7	(b) 2:1	1
8	(c) 1	1
9	(d) -25	1
10	(b) $2\sqrt{(a^2+b^2)}$	1
11	b) 42/49	1
12	(b) 1	1
13	b)6/13	1
14	(a) 60^0	1
15	a) 0.77	1
16	(a). 2 units	1
17	(c) 120°	1
18.	(a) 0	1
19	(c) Assertion (A) is true but reason (R) is false	1
20	(c) Assertion (A) is true but reason (R) is false	1
SECTION B [2 X 5 =10]		
21.	cos 60° = BASE/HYP =2.5/h 1/2=2.5/h h = 5 m. **OR** Let height of pole be h m. $\sin 30^0 = \frac{h}{20}$ $\frac{1}{2} = \frac{h}{20}$ Hence h = 10 m.	1 1 OR 1 1
22.	$(OQ)^2 = (OP)^2 + (PQ)^2$ $(12)^2 = (5)^2 + (PQ)^2$ $144 - 25 = (PQ)^2$ PQ = $\sqrt{119}\ cm.$	1 1
23	$2(1)^2 + 2(1/\sqrt{2})^2 - 2(1/\sqrt{2})^2 + (1)^2 = 3$ **OR** $4 + 4\tan^2 A$	1forall correct values 1

	$= 4 (1 + \tan^2 A)$ $= 4 (\sec^2 A) = 4 (5/2)^2 = 25$	OR 1 ½ ½
24	PE/EQ = 0.18/1.10 = 18/110 = 9/55………… **(i)** And, PF/FR = 0.36/2.20 = 36/220 = 9/55………… **(ii)** So, we get here, PE/EQ = PF/FR Hence, EF is parallel to QR	½ ½ ½ ½
25.	$54\pi = \pi x\ 36^2\ x\ \Theta/360$ $\Theta = 15$ Length of arc = $(\pi\ x\ 36\ x\ 15)/180 = 3\pi$	1 1
	SECTION C [3 X 6 = 18]	
26	$(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2$ $\sin^2 A + \operatorname{cosec}^2 A + 2 \sin A . \operatorname{cosec} A + \cos^2 A + \sec^2 A + 2 \cos A . \sec A$ $(\sin^2 A + \cos^2 A) + \operatorname{cosec}^2 A + \sec^2 A + 2 + 2$ $1 + 1 + \cot^2 A + 1 + \tan^2 A + 2 + 2$ $= 7 + \cot^2 A + \tan^2 A$ = RHS	1 for expansion 1 for conversion 1
27.	Let the fraction be x/y. According to the given information, $(x+1)/(y+1) = 1$ $x - y = -2$ …………… (1) $x/(y+1) = 1/2$ $2x - y = 1$ ………….. (2) Subtracting equation (1) from equation (2), we obtain $\boldsymbol{x = 3}$ Substituting this value in equation (1), we obtain $3 - y = -2$ $y = 3 + 2$ $\mathbf{y = 5}$ Hence, the fraction is **3/5** **OR** For infinitely many solutions $a1/a2 = b1/b2 = c1/c2$ $2/k+2 = 3/6 = -4/-(3k+2)$ $2/k+2 = 3/6$ $3k + 6 = 12$ $\boldsymbol{k = 2}$	1 for correct condition ½ for eq 1 ½ eq 2 1 OR 1 1 1
28.	Hence, we need a rod of length (in cm) which is a factor of 825, 675 and 450. For the rod to be of highest possible length we need to find the HCF of 825, 675 and 450. HCF of 825, 675, 450 825 = 5 x 5 x 3 x 11 675 = 5 x 5 x 3 x 3 x 3 450 = 2 x 3 x 3 x 5 x 5 HCF = 5 x 5 x 3 = 75 Therefore, the length of the longest rod which can measure the three dimensions of the room exactly is 75 cm **OR**	½ 1 1 for HCF ½ OR

	Correct proof	3
29	PQ+QS=8 PQ+QM=8 PS=8cm PT=8 PR+RT=8 PR+RM=8 Perimeter of triangle PQR=16cm	1 1 2
30	Modal class= 12-16, l = 12, h = 4, $f0 = 9$, $f1 = 17$, $f2 = 12$ $Mode = l + [(f1 - f0)/(2f1 - f0 - f2)] \times h = 12 + [(17 - 9)/(2 \times 17 - 9 - 12)] \times 4 = 12 + 32/13 = 12 + 2.46 = 14.46\ cm$	1 1 1
31	From $(a^2 + 9)x^2 + 13x + 6a$, Product of zeros = 1 a = 3	1 1 1
SECTION D [5 X 4 =20]		
32	Median =28.5 Median Class= 20-30 From the table, 45 + x + y = 60 x + y = 60 – 45 x + y = 15 $Median = l + (N/2 - cf)\ h/f$ $28.5 - 20 = (30 - 5 - x)/2 \quad x = 8$ $x + y = 15 \quad 8 + y = 15 \quad y = 15 - 8,\ y = 7$ $\therefore x = 8, y = 7$	Correct table 2 marks. 1 1 1
33	Statement – 1 mark Given – ½; Figure – ½; To prove- ½ Correct Proof – 2.5 marks **OR** Figure – ½ mark ; Given – ½ mark; To Prove – ½ mark; construction – ½ mark.  Proof- In ΔABD and Δ ECD AD = DE (by construction) BD = CD (D is mid point of BC) Angle ADB = angle EDC (Vert. Opp.) Hence ΔABD ≅ ΔECD (S.A.S) Hence AB = EC (CPCT) Angle BAD = angle CED (CPCT) Similarly PQ = RS {1 Marks} Angle QPM = angle RSM  $\frac{AB}{PQ} = \frac{AC}{PR} = \frac{AD}{PM} = \frac{CE}{RS}$ In ΔACE and ΔPRS $\frac{AC}{PR} = \frac{2XAD}{2XPM} = \frac{CE}{RS}$ hence $\frac{AC}{PR} = \frac{AE}{PS} = \frac{CE}{RS}$ But AB = CE and PQ = RS So ΔACE ≅ΔPRS {s.s.s} {1 mark} Hence, angle CAD = angle RPM	 1 1

	Similarly angle BAD = angle QPM Adding these results Angle BAC = Angle QPR Now In ΔABC and Δ PQR $\frac{AB}{PQ} = \frac{AC}{PR}$ and Angle BAC = Angle QPR Hence $\Delta ABC \sim \Delta PQR$ {SAS} {1 mark }	 1
34	Let the present age of the son be x years. Then, the present age of the man is ($2x^2$) years. Age of the son 8 years hence =(x+8) years. Age of the man 8 years hence ($2x^2+8$) years. ∴ $(2x^2+8)=3(x+8)+4$ $2x^2-3x-20=0 \Rightarrow 2x^2-8x+5x-20=0$ $2x(x-4)+5(x-4)=0 \Rightarrow (x-4)=0 \Rightarrow (x-4)(2x+5)=0$ $x-4=0$ or $2x+5=0$ $x=4$ or $x=-52$ $x=4$ [∵ age cannot be negative]. ∴ son's present age = 4 years, and man's present age =(2×4^2) years = 32 years.	1 1 1 1 1
35	Total volume of water overflown= (¼)×(200/3) π =(50/3)π The volume of lead shot = $(4/3)\pi r^3$ = (1/6) π Now, The number of lead shots = Total volume of water overflown/Volume of lead shot = (50/3)π/(⅙)π = (50/3)×6 = 100 OR The volume of water in the overhead tank equals the volume of the water removed from the sump. Now, the volume of water in the overhead tank (cylinder) = $\pi r^2 h$ = $3.14 \times 0.6 \times 0.6 \times 0.95\ m^3$ The volume of water in the sump when full = l × b × h = 1.57 × 1.44 × 0.95 m^3 The volume of water left in the sump after filling the tank = $[(1.57 \times 1.44 \times 0.95) - (3.14 \times 0.6 \times 0.6 \times 0.95)]\ m^3$ = $(1.57 \times 0.6 \times 0.6 \times 0.95 \times 2)\ m^3$ Height of the water left in the sump = (volume of water left in the sump)/(l × b) = (1.57× 0.6× 0.6× 0.95 ×2)/ (1.57 ×1.44) = 0.475 m = 47.5 cm Capacity of tank / Capacity of sump	1 mark 1 1 1 1 1 1 2

	= (3.14 × 0.6 × 0.6 × 0.95)/ (1.57 × 1.44 × 0.95) = 1/ 2 Therefore, the capacity of the tank is half the capacity of the sump.	1
	SECTION E [4 X 3 =12]	
36.	(1) b) 45^o (2) a) 25.24 m (3) b) 45^o **OR** b) $20/\sqrt{3}$ m	1 1 2
37	(i). Rs 5000, (ii). Production during 8th year is (a+7d)= 5000 + 7(2200) =20400 (iii). Production during first 3 year= 5000 + 7200 + 9400=21600 Or n = 12	1 1 2
38	(i) N(3,6) K(6,5) NK= $\sqrt{10}$ (ii) Akash (2,3) (iii) Midpoint (7/2, 5/2) OR Midpoint (4, 4) .	1 1 2